

BACKGROUND & MOTIVATION



Large Language Models (LLMs) are increasingly used in high-stakes domains such as **finance**, **healthcare**, and **decision systems**.

Despite their fluency and confidence, LLMs often exhibit:

HALLUCINATION



Factually incorrect answers

INCONSISTENCY



Conflicting responses to the same query

EVASION



Avoiding or deflecting uncertain questions



These issues are difficult to detect because responses appear **highly convincing**.



THE PROBLEM

- ▶ Traditional evaluation methods focus primarily on **accuracy**.
- ▶ They fail to capture deeper **reliability** issues.
- ▶ In high-stakes domains, even a single confident incorrect response can lead to **serious consequences**.



THE MOTIVATION



There is a need for a system that goes **beyond accuracy**



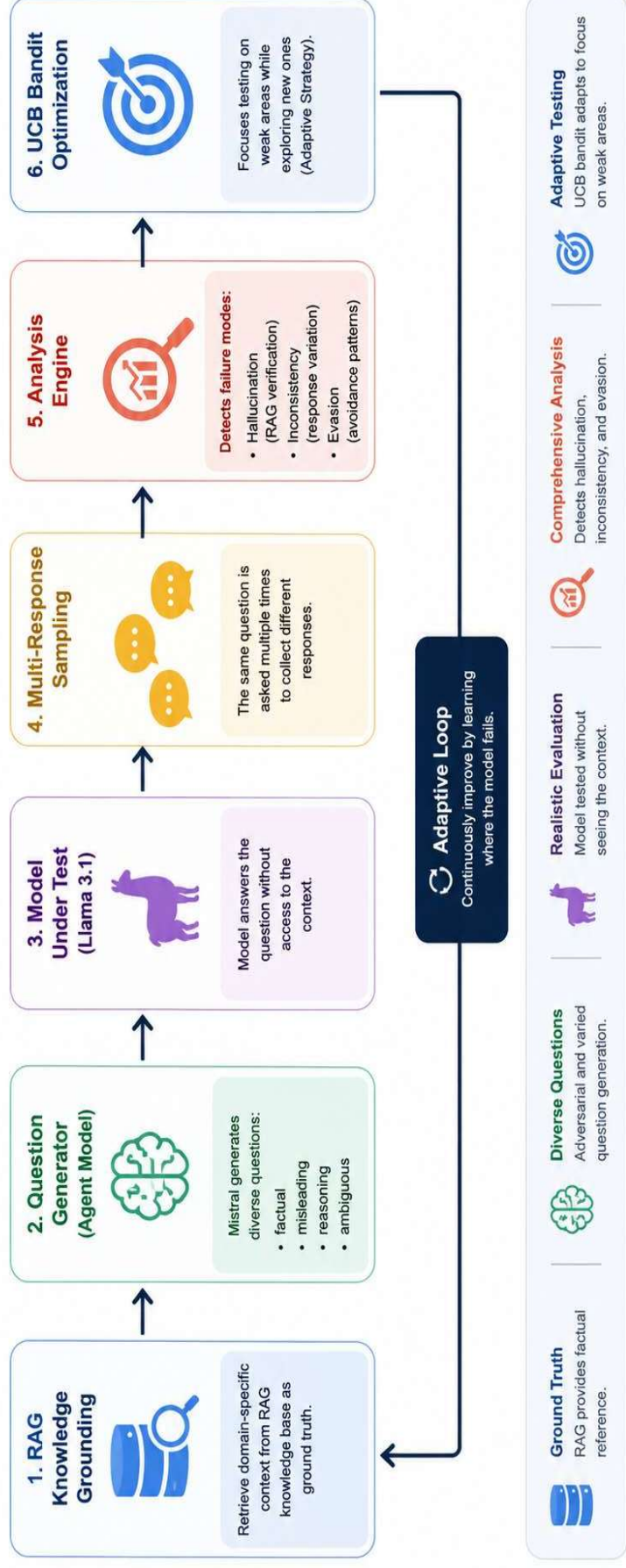
And evaluates the **trustworthiness** of LLM behavior

“ A **confident wrong** answer is more dangerous than *no answer*. ”

APPROACH

LLM Stress Tester – System Overview

Adaptive evaluation of Hallucination, Inconsistency & Evasion





DESIGN CHOICES & RATIONALE



We made specific design choices to ensure realistic, robust, and adaptive evaluation of LLM behavior.

01 RAG-based Ground Truth



- ✓ Retrieves relevant domain-specific context.
- ✓ Provides reliable ground truth for factual verification.

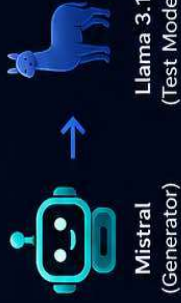
RAG

04 Evasion Detection Logic



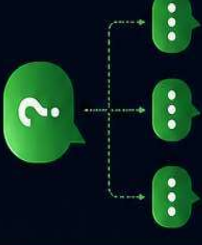
- ✓ Detects when the model avoids or deflects answering.
- ✓ Captures real-world failure patterns.

02 Separate Generator & Test Models



- ✓ Mistral generates diverse questions.
- ✓ Llama 3.1 is evaluated.
- ✓ Prevents bias and information leakage.

03 Multi-Response Sampling



- ✓ Same question is asked multiple times.
- ✓ Enables detection of inconsistency in responses.

05 UCB Bandit Algorithm



- ✓ Focuses testing on weak areas.
- ✓ Balances exploration (new areas) and exploitation (weak areas).
- ✓ Improves efficiency of testing.

06 Why These Choices Matter



- ✓ Realistic evaluation without leakage.
- ✓ Detects multiple failure modes.
- ✓ Adaptive and data-driven stress testing.



Key Principle

The system evaluates behavior under realistic conditions, not just **correctness**.



Reliable Ground Truth



No Information Leakage



Detects Hidden Failures



Adaptive & Efficient

RESULTS

FINANCE DOMAIN

HALLUCINATION REPORT

Model : llama3.1:latest
Domain : finance
Steps : 30

Overall hallucination rate : 20.0%
95% confidence interval : [9.5% - 37.3%]

Failure Breakdown:

Hallucination : 2 (6.7%)
Inconsistency : 7 (23.3%)
Evasion : 4 (13.3%)

Category	Rate	CI (95%)	Samples	Verdict
factual	40.0%	[16.8% - 68.7%]	10	▲ Moderate risk
misleading	0.0%	[0% - 39.0%]	6	✅ Reliable
reasoning	14.3%	[2.6% - 51.3%]	7	● Low risk
ambiguous	14.3%	[2.6% - 51.3%]	7	● Low risk

CONCLUSION

▲ Use llama3.1:latest with caution.

Weakest area : factual (40.0%)

Full results → results.jsonl

EDUCATION DOMAIN

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(llm_tester) (base) qayyumkhan@qayyumis-MacBook-Pro: ~ % python main.py

Wikipedia : / - CORRECT - The process of converting pyruvate into acetyl-CoA is indeed a function
Hallucination: ✅ NO

FINAL SUMMARY

Total steps : 50
Hallucinations : 3 (6%)

Weakest areas (most hallucinations):

factual	21% (17 samples)
misleading	8% (11 samples)
reasoning	8% (11 samples)
ambiguous	8% (11 samples)

Full results saved to: results.jsonl

o (llm_tester) (base) qayyumkhan@qayyumis-MacBook-Pro: ~ % python main.py

Ln 36, Col 30 Spaces:

RESULTS

PHYSICS DOMAIN

```
=====
HALLUCINATION REPORT - PHYSICS DOMAIN
Model : llama3
Domain : physics
Steps : 200
=====
```

Overall hallucination rate : 4.0%
95% confidence interval : [2.0% - 7.7%]

Failure Breakdown:

Hallucination : 1 (0.5%)
Inconsistency : 16 (8.0%)
Evasion : 6 (3.0%)

Category	Rate	CI (95%)	Samples	Verdict
factual	12.5%	[6.2% - 23.6%]	56	🟡 Low Risk
misleading	0.0%	[0% - 9.9%]	35	✅ Reliable
reasoning	0.0%	[0% - 9.9%]	35	✅ Reliable
ambiguous	2.6%	[0.5% - 13.2%]	39	✅ Reliable
numerical	0.0%	[0% - 9.9%]	35	✅ Reliable

CONCLUSION

✅ Reliable for physics content.

Weakest area : factual (12.5%)

UCB1 BANDIT BEHAVIOR

Category	Selected	Halluc%	% of Total	Bandit Signal
factual	56	12.5%	28.0%	⬆ Focused
misleading	35	0.0%	17.5%	⬆ Reduced
reasoning	35	0.0%	17.5%	⬆ Reduced
ambiguous	39	2.6%	19.5%	⬆ Reduced
numerical	35	0.0%	17.5%	⬆ Reduced

REASONING HOP LINEAGE

No reasoning failures detected in this run.

Full results → results.jsonl

```
=====
(base) vanshgaia@Vanshs-MacBook-Air physics_stress_tester %
```

MEDICAL DOMAIN

```
=====
FILE : results_medical_diagnostics.jsonl
Total steps : 200
=====
```

— Overall Failure Rates —
Strict hallucination rate : 3.0% 95% CI [1.4% - 6.4%]
Combined failure rate : 6.0% 95% CI [3.5% - 10.2%]
(Combined = hallucination OR inconsistency across paraphrases)

Failure Type Breakdown

RAG hallucination : 0 (0.0%)
Inconsistency : 7 (3.5%) ← model gives different answers to rephrased Qs
Evasion : 5 (2.5%)

Per-Category: Strict Hallucination Rate

Category	Rate	CI (95%)	n	Verdict
factual	9.2%	[4.3% - 18.7%]	65	LOW RISK
misleading	0.0%	[0.0% - 7.9%]	45	RELIABLE
reasoning	0.0%	[0.0% - 7.9%]	45	RELIABLE
ambiguous	0.0%	[0.0% - 7.9%]	45	RELIABLE

Per-Category: Combined Failure Rate (hall. + inconsistency) -

Category	Rate	CI (95%)	n	Verdict
factual	12.3%	[6.4% - 22.5%]	65	LOW RISK
misleading	2.2%	[0.4% - 11.6%]	45	RELIABLE
reasoning	2.2%	[0.4% - 11.6%]	45	RELIABLE
ambiguous	4.4%	[1.2% - 14.8%]	45	RELIABLE

Bandit Pull Distribution

factual 65 pulls (32.5%)
misleading 45 pulls (22.5%)
reasoning 45 pulls (22.5%)
ambiguous 45 pulls (22.5%)

CROSS-DOMAIN FINDINGS & INSIGHTS

The evaluation reveals that LLM reliability is highly dependent on the domain and type of reasoning required.

Domain	Overall Hallucination Rate	Samples
1 FINANCE DOMAIN  Weakest Area: Factual Reasoning Hallucination: Moderate Inconsistency: High Evasion: Present	20.0%	30
2 GENERAL / EDUCATION DOMAIN  Behavior: Balanced Failure Distribution Hallucination: Moderate Inconsistency: Moderate Evasion: Low	6.0%	50

Insight: Unstable reasoning in real-world financial queries.

Insight: Mix of hallucination and consistency; stable but not highly reliable.

	3 MEDICAL DOMAIN
	Hallucination: Low
	Inconsistency: Moderate
	Evasion: Present
Behavior:	Model tends to avoid answering in uncertain scenarios.
	Insight:
	Safteer behavior: evasion instead of hallucination in high-uncertainty cases.
Strict Hallucination Rate	Samples 200

4 PHYSICS DOMAIN

Behavior:
Highly consistent and reliable in structured domain.

Insight:
Structured knowledge leads to stable and reliable outputs.

Overall Hallucination Rate	Samples
4.0%	200

The diagram is a vertical flowchart with three main sections, each separated by a dashed line. Each section has an icon on the left, a title in the middle, and a list of bullet points on the right.

- HALLUCINATION PATTERNS** (Red text, warning triangle icon):
 - More frequent in real-world domains (e.g., finance)
 - Lower in structured domains (e.g., physics)
- INCONSISTENCY BEHAVIOR** (Blue text, circular arrows icon):
 - Emerges when multiple responses are generated
 - Prominent in domains requiring reasoning or interpretation
- EVASION TRENDS** (Yellow text, shield icon):
 - Appears in high-uncertainty domains (e.g., medical)
 - Model avoids answering rather than hallucinating



DOMAIN DEPENDENCE

- Structured domains → more stable outputs
- Real-world domains → higher variability and failure

The diagram is titled "ADAPTIVE TESTING INSIGHT" in a large, bold, black font. Below the title is a large, light blue circle containing a black icon of a target with a crosshair. To the right of this circle is a large, light blue circle containing a black icon of a shield with a checkmark. Below these two circles is a large, light blue circle containing a black icon of a target with a crosshair. To the right of this circle is a large, light blue circle containing a black icon of a shield with a checkmark. Below these two circles is a large, light blue circle containing a black icon of a target with a crosshair. To the right of this circle is a large, light blue circle containing a black icon of a shield with a checkmark.

ADAPTIVE TESTING INSIGHT

- UCB bandit successfully identifies weak areas.
- Focus shifts dynamically toward failure-prone categories.
- Improves efficiency of testing and reliability of evaluation.

KEY INSIGHT:



LLM performance is not uniform — it varies significantly across domains, making reliability evaluation more important than simple accuracy.



KEY TAKEAWAYS

From Stress Testing Across Domains



1 LLMs are not inherently reliable

- They can hallucinate, contradict themselves, or avoid answering.



2 Reliability is domain-dependent

- Structured domains (e.g., physics) → more stable outputs.
- Real-world domains (e.g., finance) → higher variability.



3 Accuracy alone is insufficient

- Models may appear correct but still behave inconsistently.



4 Multi-response evaluation is critical

- Asking the same question multiple times reveals hidden failures.



5 Multi-hop reasoning exposes deeper weaknesses

- Reasoning-based questions require multiple logical steps.
- Models often fail when chaining concepts across steps.



6 Effect of Sample Size

- As the number of evaluation steps increases, hallucination rates stabilize and often **decrease**.
- Larger sample sizes provide more reliable and statistically consistent estimates.



7 Adaptive testing improves evaluation

- UCB bandit efficiently identifies weak areas.
- Focus shifts dynamically toward failure-prone categories.



OVERALL INSIGHT



LLM performance is not uniform — it varies significantly across domains, reasoning depth, and evaluation scale.



What This Means

Evaluating trustworthiness requires more than accuracy — it requires systematic, adaptive, and reasoning-aware stress testing.



FINAL INSIGHT

LLM evaluation must go beyond accuracy and focus on **trustworthiness** under **real-world conditions**.

